

AIRTABLE_ARCHITECTURE

Base name: Securityblogs.com.au. **Base ID:** app4m600zymaqPKHX **Role in hybrid architecture:** Operational documentation + lightweight content layer (NOT primary CMS) **Primary CMS:** Payload + Postgres (see DATABASE_ARCHITECTURE)

1. Role boundary

Airtable is the **page-by-page operational reference** and the human-friendly documentation surface. It does NOT own transactional data, does NOT hold passwords, does NOT receive form submissions, does NOT serve the marketing site at request time.

Owns	Does NOT own
Per-page implementation spec (sections, copy, backend wiring)	Authentication
Documentation index linking to MD/PDF files	Leads (form submissions)
Editorial planning / rate cards / content calendars (existing tables)	Audit log
Optional: low-traffic catalogue content (logo bands, FAQ libraries)	Pages, Services, Posts (Payload owns these)
Optional: per-page status (Designed / Built / Deployed)	Subscriptions, billing, customer data

2. Current table inventory (38 tables, as of audit)

#	Table	Purpose
1	SB · 00 · Pages Index	Master index of all 36 pages
2	SB · TEMPLATE · Page	Reusable template for new pages
3	SB · 01 · Home	Per-section spec for /
4	SB · 02 · Services	/services/ hub
5	SB · 03 · Security SEO	/services/security-seo/
6	SB · 04 · AIO	/services/aio/
7	SB · 05 · AEO	/services/aeo/
8	SB · 06 · GEO	/services/geo/
9	SB · 07 · Google Ads	/services/google-ads/
10	SB · 08 · Bing Ads	/services/bing-ads/
11	SB · 09 · Web Design	/services/web-design/
12	SB · 10 · Knowledge Hub	/knowledge-hub/
13	SB · 11 · Blog	/knowledge-hub/blogs/
14	SB · 12 · Definitions & Glossary	/knowledge-hub/definitions-glossary/

#	Table	Purpose
15	SB · 13 · Industry News	/knowledge-hub/industry-news/
16	SB · 14 · Research Reports	/knowledge-hub/research-reports/
17	SB · 15 · Security Guides	/knowledge-hub/security-guides/
18	SB · 16 · Security Industry SEO (Pillar)	/knowledge-hub/security-industry-seo/
19	SB · 17 · Security Trends 2026	/knowledge-hub/security-trends-2026/
20	SB · 18 · Publish With Us	/publish-with-us/
21	SB · 19 · Advertise	/publish-with-us/advertise/
22	SB · 20 · Backlink Packages	/publish-with-us/backlink-packages/
23	SB · 21 · Guest Posting	/publish-with-us/guest-posting/
24	SB · 22 · Press Release	/publish-with-us/press-release/
25	SB · 23 · Pricing & Guidelines	/publish-with-us/pricing-guidelines/
26	SB · 24 · Product Promotion	/publish-with-us/product-promotion/
27	SB · 25 · Sponsored Posts	/publish-with-us/sponsored-posts/
28	SB · 26 · About Us	/about-us/
29	SB · 27 · Contact	/contact/
30	SB · 28 · Case Studies	/case-studies/
31	SB · 29 · Security Directory	/security-directory/
32	SB · 30 · AI Visibility Center	/ai-visibility-center/
33	SB · 31 · Free Tools	/free-tools/
34	SB · 32 · Career	/career/
35	SB · 33 · Book Strategy Call	/book-strategy-call/
36	SB · 34 · Privacy Policy	/privacy-policy/
37	SB · 35 · Terms of Service	/terms-of-service/
38	SB · 36 · Content Guidelines	/content-guidelines/

3. Per-page table schema (existing — applies to all 36 page tables)

Column	Type	Required	Purpose
Section	singleLineText (primary)	Yes	Name of the section within the page
Order	number	Yes	Top-to-bottom order on the page
Type	singleSelect	Yes	Block category (Hero / Tool/Calculator / Live demo / etc.)
Frontend – Content / Copy	multilineText	Yes	The visible text/copy in this section
Frontend – UI / Behaviour	multilineText	No	Animations, interactions, responsive behaviour
Backend – Data / Logic / APIs	multilineText	No	Implementation-grade backend reference (this doc populates it)
Build notes	multilineText	No	Editorial / build notes
Status	singleSelect	Yes	Built / Pending / In Progress / Designed

4. Proposed new documentation columns

To meet the user's spec ("Component Name, Airtable Content Source, Postgres Tables, ..."), add the following columns to each per-page table. These columns can be ADDED without touching the existing schema.

Column	Type	Notes
Component Name	singleLineText	React component or page component this section maps to
Airtable Content Source	linkedRecord → (an optional content table) OR singleLineText	If the section's content lives in another Airtable table, link it here
Postgres Tables	multipleSelects	Choose from: users, media, pages, services, case_studies, partners, posts, leads, redirects, settings, audit_log, ai_scores, ad_metrics_snapshot
API Endpoint	singleLineText	e.g. GET /api/pages?where[slug][equals]=home
Request Schema	multilineText	JSON or zod schema
Response Schema	multilineText	TypeScript shape or example JSON
Validation Rules	multilineText	Field rules + server-side checks

Column	Type	Notes
Authentication	singleSelect	Public / Bearer JWT / API Key (server) / Super Admin / Editor+
Env Vars Required	multilineText	List with one per line
Backend Service File	singleLineText	e.g. app/api/leads/route.ts:42
Frontend Component File	singleLineText	e.g. components/modules/blocks/HeroBlock.tsx
Related React Components	multilineText	Comma-separated paths
Cache Strategy	singleSelect	No cache / ISR 60s / ISR 600s / Edge cache 60s / Static at build
Revalidation Rules	multilineText	When the cache is purged (tag-based, webhook, etc.)
Deployment Notes	multilineText	Anything special at deploy time (env, migrations, secrets)
Testing Steps	multilineText	How to verify the section works
Troubleshooting	multilineText	Common failure modes + fixes
Change Procedures	multilineText	How to safely modify this section
Documentation MD	singleLineText	Relative path to Documentation/Markdown/PAGE_NAME.md
Documentation PDF	url	Link to the generated PDF

5. Documentation Index table (new — proposed)

For Airtable-discoverable documentation, create a NEW table:

Table: SB · DOCS · Index

Column	Type	Purpose
Title	singleLineText (primary)	Document name
System Name	singleSelect	Frontend / Backend / Database / Airtable / Deployment / Per-Page / Operations
Documentation Type	singleSelect	Architecture / Page Spec / Handbook / Guide / Reference / Runbook
Markdown Path	singleLineText	Repo-relative path
PDF Path	singleLineText	Repo-relative path
PDF Attachment	multipleAttachments	Optional — upload the PDF for in-base viewing
Last Updated	dateTime	Auto-stamped on import

Column	Type	Purpose
Owner	singleLineText	Engineer responsible
Linked Pages	linkedRecord → all per-page tables	Optional cross-reference

6. Sync model

Airtable is updated FROM the repo, never the other way around for transactional data.

Source of truth	Direction	Frequency
Markdown files in Documentation/Markdown/	git → Airtable (PDF attached)	On every commit to main (via GitHub Action)
Page-section implementation backend column	Engineer edits Airtable directly	Manual
Optional editorial content (low-traffic, e.g. AI engine list)	Airtable → Postgres (cached in lib/airtable.ts)	ISR 600s
Form submissions	NEVER touch Airtable	—
Auth	NEVER touch Airtable	—

7. API access

Token type	Where used	Scopes
Personal Access Token (PAT), server-side only	Marketing site lib/airtable.ts for read-only documentation lookups	data.records:read on this base only
PAT, GitHub Actions	Documentation auto-update workflow	data.records:read+write on this base only
No PAT given to browser	—	—

Rate limit: 5 req/sec per base. Cache every read in Next.js ISR with revalidate: 600.

8. Webhooks + Automations (Airtable)

Trigger	Action	Purpose
Record created in SB · DOCS · Index	Send webhook to /api/cms/purge?tag=docs	Marketing site re-fetches doc list
Status field on any per-page table changes to Built	Send webhook to GitHub Actions to regenerate PDFs	Keep docs current
New lead reported (if syncing leads READ-only from Postgres for editorial visibility)	Notify editorial in Slack	Optional

9. Risk register

Risk	Mitigation
Editor accidentally writes to Airtable expecting it to flow to production	Documentation clearly labels Airtable as docs-only; key tables read-only via Interface
Schema drift between Airtable and Postgres canonical types	Generated TS from Airtable Meta API daily; CI fails if mismatch
PAT leaked in browser	NEVER include PAT in client bundle; all reads via server route
Hitting 5 req/sec under traffic spike	Cache 600s on every read; per-IP rate limit on the read route

10. Decision log

Decision	Date	Why
Airtable used for docs, NOT primary CMS	2026-06-14	Hybrid architecture recommended in AIRTABLE_SCHEMA_MAP §10; preserves auth + leads in Postgres
Existing 36 per-page tables retained	2026-06-13	They're the natural unit of operational reference
New columns added vs. new tables	2026-06-14	Per-page tables are the natural reading unit; one row = one section = one place to look
PDF generation in repo, not Airtable Automation	2026-06-14	Deterministic; runs in CI; doesn't depend on Airtable script env

End of AIRTABLE_ARCHITECTURE.md